

Eearable Design Analysis for Sleep EEG Measurements

Swati Mandekar
IfBFH Aachen, Aachen, Germany
swati.mandekar@alumni.fh-
aachen.de

Lina Jentsch
D-HEST, ETH Zurich, Zurich,
Switzerland,
jentschl@ethz.ch

Kai Lutz
IDUN Technologies AG, Zurich,
Switzerland
kai@iduntechnologies.ch

Mehdi Behbahani
IfB, FH Aachen, Aachen, Germany
behbahani@fh-aachen.de

Mark Melnykowycz
IDUN Technologies AG, Zurich,
Switzerland
mark@iduntechnologies.ch

ABSTRACT

Conventional EEG devices cannot be used in everyday life and hence, past decade research has been focused on Ear-EEG for mobile, at-home monitoring for various applications ranging from emotion detection to sleep monitoring. As the area available for electrode contact in the ear is limited, the electrode size and location play a vital role for an Ear-EEG system. In this investigation, we present a quantitative study of ear-electrodes with two electrode sizes at different locations in a wet and dry configuration. Electrode impedance scales inversely with size and ranges from 450 k Ω to 1.29 M Ω for dry and from 22 k Ω to 42 k Ω for wet contact at 10 Hz. For any size, the location in the ear canal with the lowest impedance is ELE (Left Ear Superior), presumably due to increased contact pressure caused by the outer-ear anatomy. The results can be used to optimize signal pickup and SNR for specific applications. We demonstrate this by recording sleep spindles during sleep onset with high quality (5.27 μ Vrms).

CCS CONCEPTS

• **Human-centered computing** $\rightarrow$ Ubiquitous and mobile computing; Ubiquitous and mobile computing systems and tools.

KEYWORDS

EEG, sensors, BCI, Ear-EEG, biopotential electrodes, Sleep EEG, Impedance Spectroscopy

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1 INTRODUCTION

Electroencephalography (EEG) is a technique to measure electric potential differences produced by a group of neurons measured typically at the surface of the scalp. The EEG devices used in clinical settings are bulky, require trained personnel, and are time-consuming to setup and conduct measurements with. Hence, over the past decade, many research groups have investigated recording EEG from the outer ear canal (Ear-EEG) [1, 3, 4, 8, 14]. Ear-EEG has been shown to have similar features and quality to traditional scalp EEG [10]. The combination with a miniaturized and energy efficient biopotential front-end widened the scope to prospective uses outside of clinical and research areas. As a result, Ear-EEG appears to be the only viable choice for widespread consumer adoption of EEG technology. Applications for Ear-EEG range from brain-computer interfaces [11], work-related stress management solutions [7], and sleep monitoring [13] to cognitive-controlled hearing aids [2], epilepsy seizure warning devices [21], emotion monitoring [1] and neurofeedback home training systems [5].

As EEG records synchronized neuronal electrical activity, the signal needs to travel from the point of its origin i. e., clusters of neurons to the surface of the head through different tissue layers. For good signal acquisition, apart from the measurement device requirements to have a high input impedance and common mode rejection ratio (CMRR), the skin-electrode interface plays a crucial role. The skin contact impedance depends on the electrode's properties such as the electrode's material, size and skin conformity. Particularly for mobile applications, it is important to achieve a low-impedance contact to obtain a sufficiently high signal-to-noise ratio (SNR).

Kappel et Al. [9] investigated impedance inside the ear canal to compare wet and dry ear-EEG electrodes. They observed a mean (standard deviation) low frequency impedance (<100 Hz) of the canal electrodes of 1.2 M Ω (SD = 1.4 M Ω) and a high frequency impedance (>100 Hz) of 230 k Ω (SD = 220 k Ω) for rectangular silver electrodes of area 14 mm². The impedance measurements accounting for cerumen showed an 86% decrease in impedance after removing cerumen [18]. The electrode stabilization time in the ear before a measurement is also a major determinant of stability and low skin contact impedance, lowering impedance by up to 50% [15]. Hence, in this research, all measurements were conducted after removing the cerumen from the ear canal and after specific stabilization times.

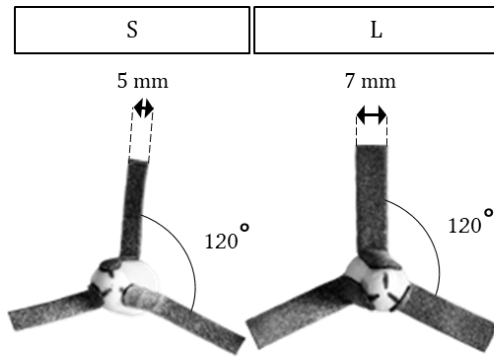


Figure 1: Foam earplug earpieces with three electrodes located at 120° apart of sizes ‘Small’ 8x5 mm (40 mm²) and ‘Large’ 8x7 mm (56 mm²)

In this study, we extend the discussion by investigating the influence of electrode size and location on the skin contact impedance. We compare the impedance for two electrode sizes and three locations inside the ear canal for dry and wet contact. Based on the findings, the optimal size and position are used to measure and characterize Ear-EEG during sleep onset. Using the proposed configuration, we were able to successfully measure a high-quality signal and extract characteristic sleep features from the measured data.

The experiments and data produced were part of the research for the Ear-EEG acquisition system for IDUN Technologies AG.

2 MATERIALS AND METHODS

2.1 Earpiece Production

A memory foam earplug is used as a base to provide more flexibility and comfort for long-duration monitoring and to adapt to the different ear canal shapes of the test subjects. The electrodes were fabricated by adding cutout silver-plated knitted conductive fabric (surface resistivity of below 5 Ω/cm^2) to the surface of the foam earplugs in order to match the mechanical deformation of the memory foam. Each earpiece comprises three electrodes oriented at an angle of 120° relative to the longitudinal axis as shown in Figure 1. Two different sizes were laser cut and bonded to the earplugs using biocompatible adhesive at 120° apart as shown in Figure 1

Each electrode was manufactured with a curvature at the tip in order to avoid skin irritation while inserting in the ear canal.

2.2 Electrode Configuration

The labeling scheme for in-ear electrode locations was adapted from P. Kidmose et Al. [10]. The electrodes are indicated by E_{xy}, where $x \in \{L, R\}$ refers to electrodes in the left or right ear, and y refers to the electrode position in the ear canal respectively. The electrode locations investigated in this study were ELE, ELJ, and ELH as shown in Figure 2

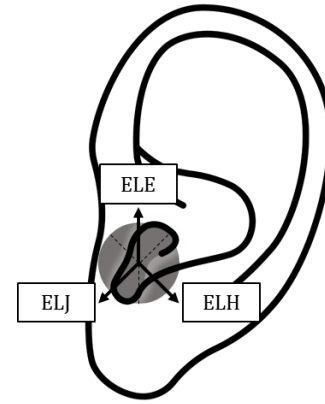


Figure 2: Labeling scheme for the in-ear electrodes, E_{xy} where $x \in \{L, R\}$ refers to the left or right ear and y is the position of the electrodes. The three positions used in this study are ELE, ELH and ELJ

2.3 Test Subjects

The measurements were performed in the left ear of 10 healthy participants after a stabilization time of 5-7 minutes. Before measurements, the participants were requested to clean their ears with a cotton swab and ethyl alcohol wipe before inserting earpieces in the ear. For dry measurements, electrodes were inserted into the cleaned but dry ear canal. For wet measurements, a drop of saline solution was applied on each electrode before the experiment. Before putting gel reference and counter electrodes on the skin, the skin was prepared by cleaning with ethyl alcohol. For hygienic concerns, each participant received their personal earpieces.

2.4 Impedance Spectroscopy

A frequency response analyzer (PalmSens4 [16]) was used to evaluate the frequency response in a three-wire impedance configuration which was connected to a Laptop (Windows Version 10) running PSTrace software [17].

The impedance measurements were carried out at the same time on all three electrodes simultaneously, whereas ELJ, ELH, and ELE served as working electrodes (WE) and standard wet gel electrodes placed at the inion as a counter (CE) and between inion and ear as reference (RE) electrodes as shown in Figure 3. Wires were crimped to the electrodes using connector pins. The medical tape was used to ensure that the electrodes outside of the ear canal did not contact each other. After the stabilization time, four frequencies were swept from 1 Hz to 100 Hz. The data is analyzed using MATLAB with mean and standard deviation.

2.5 Sleep EEG

As an initial investigation into the ability of the three electrode earpieces to detect EEG features, a 30-minute sleep session was organized. An Open BCI Cyton device was used to amplify biopotential signals from the left ear of the subject and were recorded using the OpenBCI Graphical User Interface (GUI) software package. The test subject reclined on a comfortable sofa and fell asleep

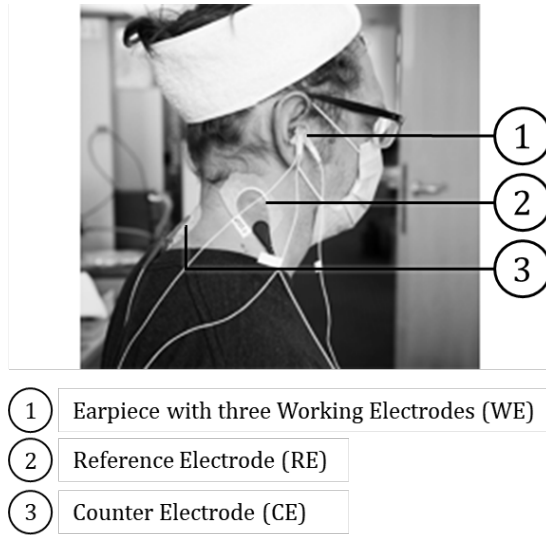


Figure 3: Electrode configuration of the ear impedance spectroscopy on the test subject for the three-electrode setup, standard wet gel electrodes are at RE and CE, WE are ear electrodes

while being monitored with a video camera to identify large body movements and aid in the results interpretation.

A stabilization time of ~10 minutes was used to allow the memory foam to stabilize in the ear. The only preparation done was cleaning ears with cotton swabs. The contra-lateral mastoid position was used for the counter and the neck was used for the reference position of the electrode setup. Standard wet gel electrodes were used as counter and reference electrodes.

3 RESULTS AND DISCUSSION

3.1 Electrode Size

The mean impedance after 5 and 7 minutes of stabilization time in the dry ear for S and L electrode sizes is shown in Figure 4 ($n=3$). The impedance was inversely proportional to the size of the electrodes and ranged from 1.25 M Ω for small to 450 k Ω for large electrodes after 5 min of stabilization time and from 1.29 M Ω (S) to 511 k Ω (L) after 7 mins of stabilization time at 10 Hz as shown in Figure 4. As reported previously in the literature, intra- and inter-subject variability can be significant for dry contact electrodes. For visual clarity, the standard deviation was excluded from the plot but can be seen in Table 1 in the supporting material.

Depending on the used amplifier, impedances should typically be well below 1 M Ω to enable high-quality signal recordings and artifact rejection [6]. At the same time, bigger electrodes mean loss of spatial resolution and number of channels. To balance this trade-off, we decided to use the smaller size for the further experiments but reduce the impedance by adding a drop of saline solution to the electrode surface prior to measurement.

In contrast to the electrode size, the two minutes difference in stabilization time did not significantly change the impedance; hence

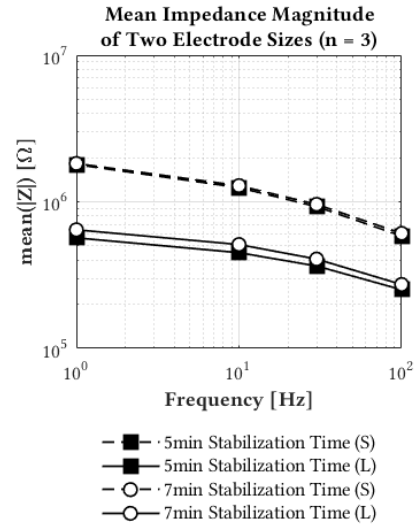


Figure 4: Mean impedance magnitude of the two electrode sizes, small (S) and large (L) at 5 and 7 mins of stabilization times, Stabilization time does not affect impedance magnitude significantly whereas the effect of electrode size on impedance magnitude can be seen clearly

a stabilization duration of 5 minutes was chosen for the following experiments.

3.2 Electrode Location

In a second experiment, impedances were compared for different positions in the ear canal using a wet skin interface. The mean impedance magnitude from 10 subjects is shown in Figure 5. The mean impedance values for small electrodes at 10 Hz were 28 k Ω (SD = 24 k Ω) for ELE, 42 k Ω (SD = 30 k Ω) for ELJ, and 29 k Ω (SD = 25 k Ω) for ELH respectively whereas for large electrodes the values were 22 k Ω (SD = 21 k Ω) for ELE, 25 k Ω (SD = 14 k Ω) for ELJ, and 29 k Ω (SD = 19 k Ω) for ELH respectively. For visual clarity, the standard deviation was excluded from the plot but can be seen in Table 2 in the supporting material.

L-sized electrodes have a lower impedance in comparison to the S-sized electrodes following the same trend as in dry measurements. The values for wet measurements for all electrodes align with the wet electrode impedance values found in the literature, 34 k Ω (SD = 37 k Ω) [9]. The irregular shape of the ear canal causes different contact pressure on the foam bud and based on anatomy, will likely be the highest at the ELE position [20]. This is also reflected in the impedance results for both sized and contact conditions. Larger electrodes could average out local differences in ear canal surface and the differences between locations becomes smaller. Additionally, the use of ionic solution covering the electrode surface area can reduce the effect of size areas as a liquid film can compensate for skin irregularities. However, fluid might cause issues related to cross-talk between electrodes and hence, there needs to be a tradeoff between size and number of channels. So, the sleep EEG

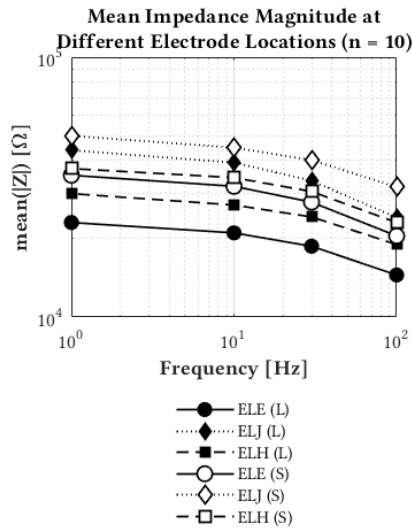


Figure 5: Mean impedance magnitude of the two electrode sizes small (S) and large (L) at locations ELE, ELJ and ELH. Among two sizes, Location ELE shows the lowest impedance compared to the other two locations of both sizes.

measurements were performed with small size electrode earpieces, ELE as measurement electrode.

3.3 Sleep EEG

The raw signal from the sleep trial was visually investigated using the OpenBCI GUI. Of particular interest were large-signal deviations resulting from subject movements (motion artefacts) as well as signal changes related to the physiological sleep state. Common sleep features include spindles and the slow-wave k-complex. These are seen as bursts in the EEG signal and occur close to one another. The data was filtered with a notch filter at 50 Hz to address line noise as well as a band-pass filter from 1 to 50Hz. It was observed that the Ear-EEG signal was stable and bursts of activity with associated slow waves were present. The signal characteristics are similar to spindle activity from Ear-EEG reported by Mikkelsen et. Al [12]. This provided the first evidence that sleep features could be detected using the Ear- EEG electrode setup.

4 CONCLUSION

This study showed the impact of electrode size and location on the skin contact impedance inside the ear-canal for both dry and wet skin contact. Impedances ranged from 450 kΩ to 1.29 MΩ for dry and from 22 kΩ to 42 kΩ for wet contact at 10 Hz and was inversely proportional to the electrode area. The location ELE provided the lowest impedance irrespective of size, likely due to increased attachment pressure to the skin resulting from the anatomy of the ear canal.

We combined small electrode size and the ELE location to detect sleep onset from the ear canal. Characteristic sleep spindles were clearly distinguishable from the baseline during the measurement

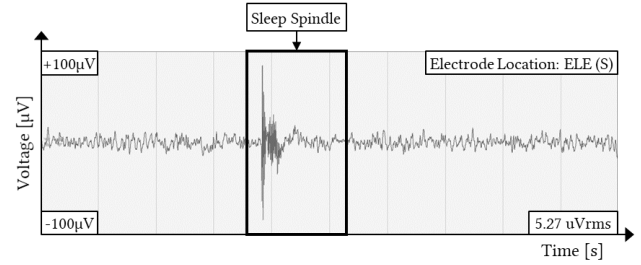


Figure 6: Sleep-EEG signal recorded from the ear during the sleep trial. Bursts in the signal were present and reminiscent of sleep spindle activity as seen in the literature.

The ability to measure sleep activity is ideal for soft electrode designs, as the material can conform will to the ear canal anatomy. During sleep subject movements will be small to minimize motion signal artefacts.

The next step in evaluating soft electrodes includes a comparison to scalp-based EEG to look more closely at correlations between the scalp and ear canal locations and the resulting signal quality.

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